# rag -- 2026-11-10

*5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.*

## Worth reading in full

**[P1] PCG-RAG: Progressive Compositional Grounding for Retrieval-Augmented Generation in Complex Web Question Answering** -- PCG-RAG offers the most comprehensive advances in grounding, compositional retrieval, and empirical gains across multiple complex Web QA benchmarks, making it essential reading for anyone studying RAG system design.

## Themes

Several papers focus on improving retrieval quality for multi-hop QA by redesigning the retrieval-generation pipeline. PCG-RAG introduces progressive evidence acquisition and compositional reasoning that treat partial entity matches as revisable hypotheses, yielding up to 71.33 % accuracy on BrowseComp-Plus and 5-11 point gains over strong baselines such as WebThinker and LQT [P1]. Lightfoot builds a deterministic graph index that requires no model calls at retrieval time and supports exact, model-free document deletion, is 26-31 % cheaper to index, has a median retrieval-only latency of 0.53 s, improves recall on MuSiQue, improves exact-match on HotpotQA, and shows a drop in recall on 2WikiMultiHopQA [P3].

Robustness to noisy retrievals is addressed by ARMOR, which decomposes RAG robustness into three sub-tasks and uses an adaptive curriculum meta-learning scheduler; it maintains 69 % correctness even when 80 % of retrieved documents are noisy, outperforming prior methods by up to 38.8 % on correctness scores [P2]. By contrast, Yuanbao-Reranker concentrates on the reranking stage, synthesising document-grounded training data and applying a minimum-gap hinge loss to achieve state-of-the-art nDCG@10 across multilingual and code retrieval benchmarks, but it does not directly modify the retrieval or generation components [P5].

Evaluation breadth varies: PCG-RAG and ARMOR report results on several multi-hop QA benchmarks (BrowseComp-Plus, WebWalkerQA, MuSiQue, HotpotQA, 2WikiMultiHopQA), while Lightfoot adds cost and latency analyses. MMGraph-Agent extends the discussion to multimodal RAG, constructing a zero-API hypergraph and a cache-inspired memory to handle long contexts across 23 multimodal QA datasets, though numeric gains are not disclosed [P4]. The collection therefore highlights three complementary research strands--progressive grounding, graph-based cost-efficient retrieval, and noise-robust curriculum learning--each tackling different facets of retrieval-augmented generation design and evaluation.

*5 theme sentence(s) checked against the paper summaries; 1 corrected, 0 removed.*

## At a glance

**[P1] PCG-RAG: Progressive Compositional Grounding for Retrieval-Augmented Generation in Complex Web Question Answering** · relevance 0.49
PCG-RAG is a retrieval-augmented generation framework that progressively grounds entity identification and compositional reasoning, reaching up to 71 % accuracy on complex Web QA and surpassing strong baselines by 5-11 percentage points.

**[P2] ARMOR: Adaptive Curriculum Meta-Learning for Noise-Robust RAG Reasoning** · relevance 0.45
ARMOR introduces an adaptive curriculum meta-learning framework that markedly improves noise-robustness of retrieval-augmented generation on multi-hop QA datasets.

**[P3] Lightfoot: Graph Retrieval-Augmented Generation with Exact, Model-Free Document Deletion** · relevance 0.44
Lightfoot is a GraphRAG system that builds a cheap, deterministic index with no model calls at retrieval time and supports exact, model-free document deletion, achieving competitive multi-hop QA performance but lagging on one benchmark.

**[P4] MMGraph-Agent: Agentic Multimodal RAG via Cache-Inspired Multimodal Knowledge HyperGraphs** · relevance 0.43
MMGraph-Agent is a unified multimodal retrieval-augmented generation system that builds a zero-API-cost hypergraph and uses a cache-inspired memory to improve efficiency and handle long contexts, showing consistent gains over baselines on 23 multimodal QA datasets.

**[P5] Yuanbao-Reranker: Document-Grounded Synthesis and Preference Distillation for Text Reranking** · relevance 0.39
Yuanbao-Reranker introduces dense, document-grounded rerankers (0.6B-8B) that achieve state-of-the-art nDCG@10 scores across Chinese, multilingual, English, long-document, and code retrieval benchmarks.

## [P1] PCG-RAG: Progressive Compositional Grounding for Retrieval-Augmented Generation in Complex Web Question Answering

*Linyao Zheng, Xuhang Shi, Zhifang Mao, Chongqi Qu et al. · arXiv 2611.11113v1 · https://arxiv.org/abs/2611.11113v1 · relevance 0.49*

**TL;DR:** PCG-RAG is a retrieval-augmented generation framework that progressively grounds entity identification and compositional reasoning, reaching up to 71 % accuracy on complex Web QA and surpassing strong baselines by 5-11 percentage points.

**Problem**
Complex Web question answering requires systems to locate target entities from multi-clue descriptions and to stitch together evidence that is scattered across long Web pages, where individual pages often satisfy only a subset of the needed constraints and premature reliance on partially supported hypotheses can misdirect retrieval [1][2].

**Method**
PCG-RAG introduces three tightly coupled components: (1) Progressive Evidence Acquisition, which treats partial entity matches as revisable search hypotheses to retrieve missing identifying evidence [2][3]; (2) Compositional Reasoning, which separates candidate-guided retrieval from evidence-based answering by traversing a directed acyclic graph of subquestions and instantiating dependent queries [3]; and (3) Evidence Grounding, which merges identifying and relational source passages with intermediate answers to produce a final generation grounded in the original question [2][7].

**Key results**
- On the BrowseComp-Plus benchmark PCG-RAG attains 71.33 % answer accuracy, exceeding the WebThinker baseline by 5.43 points [5] and surpassing the best RAG baseline (LQT) by 11.33 points [5].
- On WebWalkerQA it achieves 64.56 % accuracy, 5.15 points higher than DeepHone and 7.80 points above LQT [5].
- The abstract also reports the same improvements of 5.43 and 5.15 percentage points over the leading baselines on the two datasets [1].
- In a reasoning-depth analysis on MuSiQue, PCG-RAG reaches All@5 scores of 84.17 % (2-hop), 56.65 % (3-hop), and 16.87 % (4-hop), outperforming comparable methods such as MGranRAG, CIRAG, RADAR, and LQT [6].

**Limitations**
The provided excerpts do not discuss any limitations of PCG-RAG [7].

***Sources***

[1] arXiv 2611.11113v1.pdf, arXiv abstract

[2] arXiv 2611.11113v1.pdf, page 2

[3] arXiv 2611.11113v1.pdf, page 4

[5] arXiv 2611.11113v1.pdf, page 5

[6] arXiv 2611.11113v1.pdf, page 7

[7] arXiv 2611.11113v1.pdf, page 8

## [P2] ARMOR: Adaptive Curriculum Meta-Learning for Noise-Robust RAG Reasoning

*Yan Wang, Yuxin Zhang, Shenyu Zhang, Yongrui Chen et al. · arXiv 2611.11771v1 · https://arxiv.org/abs/2611.11771v1 · relevance 0.45*

**TL;DR:** ARMOR introduces an adaptive curriculum meta-learning framework that markedly improves noise-robustness of retrieval-augmented generation on multi-hop QA datasets.

**Problem**
Retrieval-augmented generation (RAG) systems suffer from hallucinations caused by noisy retrieved documents, and existing approaches address only isolated aspects of robustness such as identification, filtering, or reasoning rather than the full composite challenge [1].

**Method**
ARMOR decomposes RAG robustness into three sub-tasks--knowledge boundary perception, retrieval verification, and robust reasoning--and employs an Adaptive Task Scheduler that dynamically selects the most beneficial sub-task for each training batch based on the generator's Absolute Learning Progress [2]. It also introduces Inter-Group Advantage Estimation, which computes reward advantages across the global batch to prevent mode collapse and stabilize reinforcement-learning optimization [2][5].

**Key results**
On three multi-hop QA benchmarks (2WikiMultiHopQA, MusiQue, HotPotQA), ARMOR achieves state-of-the-art correctness, F1, and exact-match scores with both Qwen-3-1.7B and Llama-3.2-3B backbones, e.g., 79.14 % Correctness on 2WikiMultiHopQA with Qwen-3-1.7B (F1 79.22 %, EM 77.22 %) and 77.81 % Correctness with Llama-3.2-3B (F1 78.09 %, EM 75.59 %) [4].
Compared to the strongest baseline IRCOT, ARMOR improves average correctness by 38.8 % on Qwen-3-1.7B and 19.9 % on Llama-3.2-3B [3].
In a controlled noise-ratio experiment, ARMOR maintains a correctness of 69 % even when 80 % of retrieved documents are noisy, far outperforming baselines that collapse under high noise [4].

**Limitations**
The evaluation is limited to general-domain English multi-hop QA datasets (2WikiMultiHopQA, MuSiQue, HotpotQA) and does not assess performance on specialized or high-stakes domains such as mathematics or healthcare [6].
Additionally, the experiments are confined to English corpora, leaving multilingual robustness unexamined [6].

***Sources***

[1] arXiv 2611.11771v1.pdf, arXiv abstract

[2] arXiv 2611.11771v1.pdf, page 3

[3] arXiv 2611.11771v1.pdf, page 5

[4] arXiv 2611.11771v1.pdf, page 6

[5] arXiv 2611.11771v1.pdf, page 7

[6] arXiv 2611.11771v1.pdf, page 8

## [P3] Lightfoot: Graph Retrieval-Augmented Generation with Exact, Model-Free Document Deletion

*Uday Allu · arXiv 2611.11010v1 · https://arxiv.org/abs/2611.11010v1 · relevance 0.44*

**TL;DR:** Lightfoot is a GraphRAG system that builds a cheap, deterministic index with no model calls at retrieval time and supports exact, model-free document deletion, achieving competitive multi-hop QA performance but lagging on one benchmark.

**Problem**
Graph-based retrieval-augmented generation (GraphRAG) improves multi-hop question answering but deployed systems incur two under-measured costs: the expense of building the graph and the difficulty of removing documents from it [1][2]. Existing open-source frameworks rarely offer exact deletion, and none guarantee that the resulting graph matches one built without the deleted documents [2].

**Method**
Lightfoot constructs a graph where every entity, link, fact, and synonym edge is a deterministic function of per-passage extractions that retain chunk-level provenance [1][3]. Indexing requires a single language-model call per passage, retrieval makes no model calls, and document deletion is performed as a pure database operation with no model calls [1][3]. Retrieval seeds a Personalized PageRank walk with the two most similar facts, adds the 100 most similar passages, runs the walk (damping 0.9), selects the top 30 passages, and fuses their scores with a local cross-encoder rank (reciprocal rank fusion k = 60) [3]. The answer model receives the ten highest-scoring facts and the top 10 passages and returns a short answer span; no language model is invoked before this step [3].

**Key results**
- Index cost: Lightfoot is 26-31 % cheaper than HippoRAG 2 and 3-6 × cheaper than LightRAG and fast-graphrag across MuSiQue, 2WikiMultiHopQA, and HotpotQA [1][5][6].
- Retrieval performance (vs. HippoRAG 2): on MuSiQue, recall@5 improves by +2.8 points and recall@10 by +4.3 points; on HotpotQA, exact-match (EM) improves by +7.7 points; on 2WikiMultiHopQA, recall@2 drops by -3.9 points and F1 by -8.4 points [1][5].
- Detailed paired differences (Table 2) show MuSiQue recall@5 +2.8 [+0.5, +5.1], recall@10 +4.3 [+2.5, +6.2]; HotpotQA EM +7.7 [+3.0, +12.3]; 2WikiMultiHopQA recall@2 -3.9 [-7.0, -0.8] and F1 -8.4 [-12.9, -4.2] [5].
- Model-call cost per passage: Lightfoot uses 1.0 call (~ $1.64 for MuSiQue), HippoRAG 2 uses 2.0 calls (~ $2.23), fast-graphrag 1.1 calls (~ $4.89), LightRAG ~ 2.5 calls (~ $9.5) [6].
- Retrieval latency (median, retrieval-only): Lightfoot 0.53 s, LightRAG 0.81 s, HippoRAG 2 1.56 s, fast-graphrag 1.96 s [5].
- Deletion: Lightfoot deletes a seeded 10 % of MuSiQue documents with zero model calls; LightRAG requires 2,187 model calls and 3.7 hours; fast-graphrag lacks a deletion API [5][6].

**Limitations**
Lightfoot shows that a GraphRAG index can be cheap to build, need no model at retrieval time, and support exact, model-free deletion, while matching the strongest open-source baseline on multi-hop accuracy overall, with the remaining accuracy gap attributed to selecting seed facts and reaching later hops that the question never names [7]. Deletion audits show that recall on questions whose evidence was removed drops sharply, confirming dependence on the deleted content [5]. Fast-graphrag does not provide a model-free deletion path, and Lightfoot omits a model-based name-pair judge whose removal changed recall@5 by less than one point [3][6].

***Sources***

[1] arXiv 2611.11010v1.pdf, arXiv abstract

[2] arXiv 2611.11010v1.pdf, page 1

[3] arXiv 2611.11010v1.pdf, page 3

[5] arXiv 2611.11010v1.pdf, page 5

[6] arXiv 2611.11010v1.pdf, page 6

[7] arXiv 2611.11010v1.pdf, page 7

## [P4] MMGraph-Agent: Agentic Multimodal RAG via Cache-Inspired Multimodal Knowledge HyperGraphs

*Haoran Luo, Ziyue Zhu, Shangyang Wu, Linhao Luo et al. · arXiv 2611.10657v1 · https://arxiv.org/abs/2611.10657v1 · relevance 0.43*

**TL;DR:** MMGraph-Agent is a unified multimodal retrieval-augmented generation system that builds a zero-API-cost hypergraph and uses a cache-inspired memory to improve efficiency and handle long contexts, showing consistent gains over baselines on 23 multimodal QA datasets.

**Problem** - Current multimodal RAG pipelines suffer from expensive knowledge organization, limited context windows, and a split between offline and online knowledge sources, which hampers scalability and unified training [1].

**Method** - MMGraph-Agent constructs a multimodal knowledge hypergraph with no external API cost and employs a two-tier (L1/L2) cache-inspired memory that bounds the prompt length via digest-bounded admission (C = 5, Lmax = 1,200 words, ~300-word digest, Lcap = 5,000 tokens) [2]. A single policy ??, trained offline with multimodal reinforcement learning (GRPO) on 8 × A100 GPUs for 75 steps (G = 8, ? = 0.2, ? = 10?[3]), selects text-query, image-query, or answer actions in a unified action space and is deployed online without further updates [2][6].

**Key results** - The system is evaluated on 23 multimodal QA datasets spanning knowledge-only, web-search-only, and combined regimes [3][4]. Experiments report that MMGraph-Agent consistently outperforms generation-only, RAG-based, and training-based baselines, and ablations confirm the complementary contribution of its three modules [4]. No specific numeric scores or percentage gains are provided in the excerpts. Training details include 75 optimization steps with batch size G = 8 and hardware of 8 A100 GPUs [2].

The authors note that the offline hypergraph requires a one-time construction cost, web-only datasets rely on an external search API, and all experiments use a single backbone scale, with other scales left to future work [7].

***Sources***

[1] arXiv 2611.10657v1.pdf, arXiv abstract

[2] arXiv 2611.10657v1.pdf, page 5

[3] arXiv 2611.10657v1.pdf, page 6

[4] arXiv 2611.10657v1.pdf, page 9

[6] arXiv 2611.10657v1.pdf, page 21

[7] arXiv 2611.10657v1.pdf, page 27

## [P5] Yuanbao-Reranker: Document-Grounded Synthesis and Preference Distillation for Text Reranking

*Hongyuan Xu, Dezhi Ye, Jiahao Cao, Jie Liu et al. · arXiv 2611.10405v1 · https://arxiv.org/abs/2611.10405v1 · relevance 0.39*

**TL;DR:** Yuanbao-Reranker introduces dense, document-grounded rerankers (0.6B-8B) that achieve state-of-the-art nDCG@10 scores across Chinese, multilingual, English, long-document, and code retrieval benchmarks.

High-precision text reranking is a critical component in modern search, recommendation, and retrieval-augmented generation systems [1].

**Method** - The authors synthesize query-positive pairs via answer-first and query-type-guided document-grounded synthesis, then apply a three-stage curation funnel (teacher verification, deduplication, source-length quotas) to build 2.75 M 1:3 ranking groups containing hard, medium, and moderate negatives [2][3]. Models (0.6B, 4B, 8B) are fine-tuned with pointwise relevance cross-entropy and a bounded minimum-gap hinge loss over positive-negative and negative-negative pairs, followed by layerwise adaptive model merging [2][4].

On the shared Qwen Top-100 candidate pool, the 0.6B model attains an average nDCG@10 of 70.96, surpassing all sub-1B baselines and outperforming 1B-4B baselines outside the Qwen3 family [5]. The 4B model reaches 73.63 Avg., beating Qwen3-Reranker-4B (72.13) and the 8B teacher (72.68) [5]. . Ablation studies show that the full minimum-gap hinge objective improves overall Avg. by +1.23 points over SFT-only training, with per-benchmark gains up to +1.54 on long-document retrieval (MLDR) [6].

**Limitations** - The excerpts do not provide an explicit discussion of the paper's limitations; such analysis is absent from the available text.

***Sources***

[1] arXiv 2611.10405v1.pdf, arXiv abstract

[2] arXiv 2611.10405v1.pdf, page 3

[3] arXiv 2611.10405v1.pdf, page 7

[4] arXiv 2611.10405v1.pdf, page 10

[5] arXiv 2611.10405v1.pdf, page 11

[6] arXiv 2611.10405v1.pdf, page 13